

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

Course Name: CSA1521 - Cloud
Computing and Big Data Analytics.

Name: Yogashree G

Reg No: 192573022

Date: 04/08/2026

CO2-AT1 - Analytical Problem Solving

Q1) Compare Type-1 and Type 2 Hypervisors. Justify.

Type 1 - Hypervisor

- * Runs directly on the hardware without a host OS.
- * Provides high performance with low overhead.
- * Offers better security because there is no host operating system to attack.
- * Easy to manage in large cloud datacenters.

Type 2 - Hypervisor

- * Runs on top of a host operating system.
- * Has higher resource overhead and lower performance.
- * Less secure because attacks on the host OS can affect VMs.
- * Suitable for testing, development, and personal use.

Justification:

For mission-critical cloud applications, Type 1 Hypervisor is the better choice because it provides better performance, stronger security, higher availability, and easier management for large scale cloud environments.

Comparison:

<u>Feature</u>	<u>Type-1</u>	<u>Type-2</u>
Performance	High	Moderate
Security	High	Lower
Manageability	Excellent	Limited
Best Use	Cloud Datacenters	Development & Testing.

Q6) Virtualization Architecture for Workload Isolation.

Virtualization allows multiple workloads such as finance, healthcare, and student portals to run securely on the same physical server.

- * Each workload runs inside a separate Virtual Machine (VM).

- * The Hypervisor Isolates CPU, memory, storage and network resources.

- * A failure or attack in one VM does not affect other VMs.

- * VM separates traffic between applications.

Major Attack Surface:

- 1) Hypervisor Attack - If the hypervisor is compromised,

all hosted VMs may be affected.

2) VM Escape Attack - A malicious VM breaks isolation and gains access to the host or other VMs.

Conclusion:

Virtualization provides secure workload isolation while maximizing hardware utilization. Strong hypervisor security and regular updates reduce attack risks.

Q3) Analyze the VM host snapshot.

→ CPU = 92%

→ Memory = 88%

→ Storage I/O latency = 35 ms

→ VM Boot Time = 170 s.

Identify Bottlenecks.

- * CPU usage is very high, causing slower processing.
- * Memory utilization is close to maximum, leading to performance degradation.
- * Storage latency is high, delaying data access.
- * Long VM boot time indicates overloaded compute and storage resources.

Corrective Actions:

- Migrate some VMs to another host.
- Increase CPU and RAM capacity.
- Replace HDD with SSD/NVMe storage.
- Enable load balancing and optimize resource allocation.
- Monitor resource usage continuously.

Conclusion.

The host is overloaded in CPU, memory, and storage. Resource optimization and VM migration will improve overall performance.

Q4) Software Defined Datacenter (SDDC) vs Traditional Datacenter:

A software Defined Datacenters (SDDC) manages compute, storage, and networking software instead of manual hardware configuration.

Advantages:

- Faster deployment of resources
- Automated management and scaling
- Better resource utilization
- Lower operational cost.
- Higher flexibility and agility.

Virtualization Components:

- (i) Compute Virtualization : Creates multiple VMs on a single server.
- (ii) Storage Virtualization: Combines multiple storage devices into one logical pool.
- (iii) Network Virtualization : Creates virtual networks with software - defined control.

Conclusion:

SDN improves agility by automating infrastructure management, enabling faster provisioning, efficient scaling, and simplified cloud operations.

Q5) Multi-Cloud Federation Identity Mismatch :

Different cloud providers use different identity systems, causing authentication failures and inconsistent access control.

Issues:

- 1) User Identity mismatch
- 2) Multiple login credentials.
- 3) Difficult access Management.
- 4) Increased security risks.

Secure Identity mismatch:

- ↳ Use Federated Identity Management for Single Sign-On.
- ↳ Implement centralized Identity provider (IdP).
- ↳ Use secure authentication protocols such as SAML or OAuth/Open ID connect.
- ↳ Apply Role-Based Access Control (RBAC).

Conclusion:

Federated Identity provides secure authentication, simplifies user access across multiple cloud providers, and protects user privacy while reducing identity management complexity.